

GeoMatch: photo geolocation as visual retrieval

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1 Problem

Predict the GPS coordinate of a photo from twelve European countries, scored by median haversine error. The rules allow *one* model of at most 5,000,000 parameters, no pretrained weights, and no data beyond the 11,758 labelled training images.

Regression and grid classification both quantise the answer, so the model treats geolocation as **retrieval**: embed the query and every training image into a common space, take the nearest training images, and read the coordinate of the best match. This yields a continuous prediction from a compact encoder.

2 Architecture (Fig. 1)

A single RegNetY-400MF trunk (`weights=None`) is shared across three views: the full frame at 512^2 and aligned left/right crops at 256^2 , which retain high-resolution detail a single downsampled frame loses. Two branches read it:

Global. Multi-scale GeM pooling over stages 2–4, fused across views, passed through an identity-initialised 384×384 layer and L2-normalised to a 384-D descriptor. Cosine similarity to the bank gives a top-200 shortlist.

Local. Stage-4 features average-pooled to a 4×4 grid per view (48 tokens), projected $440 \rightarrow 64$, L2-normalised. A bidirectional MaxSim (ColBERT-style late interaction) reranks the shortlist.

Final score = $0.2 \text{ global} + 0.8 \text{ local}$; the top-1 candidate’s coordinate is the prediction. Country and geo-cell heads on the fused descriptor act only as a training regulariser.

3 Training

Two stages, evaluated on the official 5-fold split (train four, predict the fifth, fixed epochs, no checkpoint selection).

1. Self-supervised pretraining. BYOL (negative-free, momentum target and predictor), labels unused, 160 epochs. A trunk warm-started only by the classification signal never learns features that separate locations *within* a large, visually uniform country; BYOL gives it a sharper starting representation.

2. Country-aware retrieval fine-tuning, 40 epochs from the BYOL trunk. The loss combines spatial-listwise, local-listwise, triplet and local-verification terms. Each group is an anchor, two geographic positives (≤ 35 km) and one *same-country* hard negative mined from the current EMA encoder and refreshed every epoch, forcing the encoder to separate nearby same-country scenes. Anchors from the weakest countries are oversampled.

4 Results

	median (km)	≤ 50 km
From-scratch retrieval baseline	89.2	45.3%
+ BYOL + country-aware finetune	55.6	49.1%

Table 1: Pooled 5-fold out-of-fold error. Three seeds span 55.6–57.7 km. The submitted model retrains this recipe on all 11,758 images.

Every country improves (Fig. 2): Iceland 2.9 km, Norway 9.1, Finland 15, with GB (63) and SE (57) near the 50 km line. Longer schedules and a finer 8×8 local grid both overfit and hurt; augmenting the global view discarded scene context and hurt.

The gap is ranking, not recall (Fig. 3). A within-50 km training image is in the top-200 for $\sim 80\%$ of queries, but the decoder returns a within-50 km match only 49% of the time. The right answer is retrieved and then mis-ranked.

5 Germany and France: unsolved

Every other country’s median is under 220 km; Germany (505) and France (570) do not move. This is a representation gap, confirmed three ways: retrieval ranking, retrieval recall (DE/FR oracle@200 plateaus near 60% vs. 80% pooled), and the geo-cell head, which is equally lost there. Country-aware oversampling and same-country hard negatives target exactly this and barely helped.

German and French street scenes—suburban architecture, road furniture, vegetation—repeat across hundreds of kilometres, while Nordic and Icelandic scenes carry distinctive terrain. A $\leq 5\text{M}$ -parameter from-scratch encoder at 512 px lacks the capacity to read the fine localised cues (signage, plates, micro-architecture) that separate them. That is the clearest direction for a larger budget.

6 Submission

The submitted model retrains the recipe once on all 11,758 labelled images (no held-out fold), from an existing BYOL trunk: 4,869,911 parameters, ≈ 70 min on one RTX 4000 Ada, plus ≈ 5 min to write `predictions.csv` (2,400 rows) via the top-200 shortlist and late-interaction rerank.

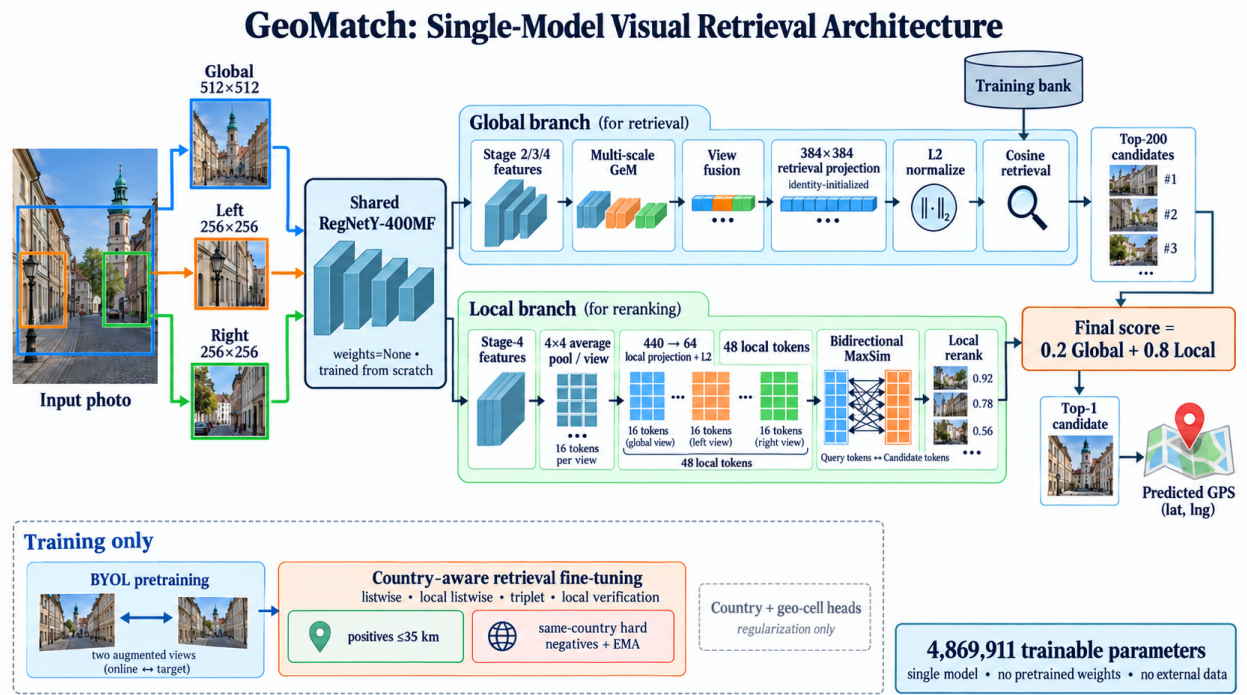


Figure 1: The full pipeline. One from-scratch RegNetY-400MF trunk, shared across a global view and two crops, feeds a global-descriptor retrieval branch and a local-token late-interaction reranker. Classification heads regularise training only. 4,869,911 parameters.

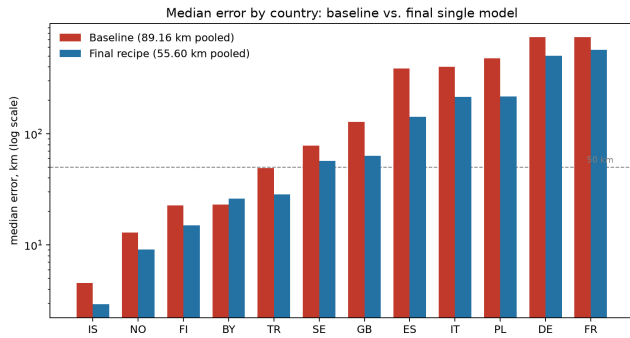


Figure 2: Median error by country. Every country improves except Germany and France.

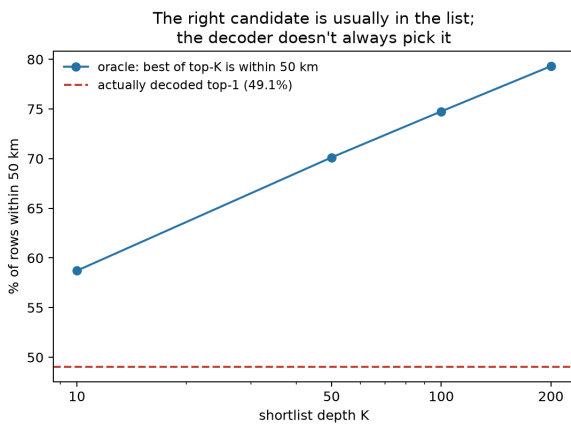


Figure 3: Oracle (best of top- K within 50 km) vs. the decoded top-1.

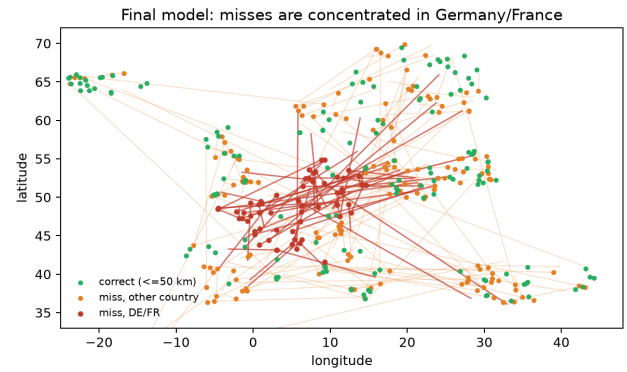


Figure 4: Misses (lines) are concentrated in Germany/France (red), not spread.